Advancing Bi-objective Weighted Portfolio Optimization Across International Markets via Quantum Annealing

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2025 IEEE International Conference on Quantum Computing and Engineering (QCE) | 979-8-3315-5736-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/QCE65121.2025.00245

***Abstract*—Quantum finance has emerged as a key research area as interest grows in leveraging quantum computing for solving complex optimization problems in financial technology. Quantum annealing (QA), which leverages quantum tunneling effects, presents a promising method for tackling portfolio optimization challenges that primarily involve balancing risk and return. Traditional single-solution optimization approaches often fail to accommodate diverse investor preferences regarding risk tolerance and expected returns. To bridge this gap, we propose a novel bi-objective weighted portfolio optimization framework based on the trend ratio (TR) model. Our approach integrates D-Wave's advanced constrained quadratic model (CQM) hybrid solver with QA and employs scalarization-based non-dominated analysis, enabling multiple runs with different weights between risk and return to effectively achieve bi-objective optimization. This flexible strategy generates a diverse set of high-quality solutions tailored to different investor profiles. In our experiments, we investigate the interaction between two well-known international stock indices, DJI and Nikkei, to gain insights into cross-market dynamics and enhance portfolio diversification. Experimental results show that the proposed approach consistently demonstrates strong potential compared to conventional multi-objective optimization algorithms, delivering solutions with improved quality and computational efficiency. These findings highlight the potential of quantum computing in advancing financial decision-making and portfolio optimization.**

***Index Terms*—Quantum Finance, Quantum Computing, Bi-objective Optimization, Weighted Portfolio Optimization**

I. Introduction

Quantum finance has emerged as a promising interdisciplinary field, leveraging the computational capabilities of quantum computing to address complex problems in financial decision-making. Among these, portfolio optimization [1], [2] is a fundamental yet challenging task, as it involves handling conflicting objectives, namely minimizing risk and maximizing expected return, while accounting for the intricate interactions among assets. Traditional optimization methods often struggle to explore the vast solution space efficiently and to accommodate diverse investor preferences.

Quantum-inspired optimization (QIO) [3]–[5] and quantum annealing (QA) have both shown strong potential for solving complex optimization problems. Among them, QA [6], [7] offers strong potential by leveraging quantum tunneling to escape local optima and explore more promising regions of the solution landscape. D-Wave's QA systems [8], which include specialized quantum hardware and hybrid solvers, have demonstrated promising results in a variety of domains, such as sensor placement [9], electric mobility [10], drug design [11], quadratic assignment [12], traffic assignment [13], and portfolio optimization [14]. However, most existing QA applications in finance focus on single-objective formulations. The application of QA to multi-objective portfolio optimization, particularly where both risk and return are jointly optimized, remains largely unexplored.

Portfolio optimization [15] aims to determine the most effective asset allocation strategy that maximizes return while minimizing risk. In practice, investor preferences toward the risk–return trade-off vary significantly. Some seek aggressive returns with higher risk exposure, while others adopt conservative strategies with lower risk tolerance. To accommodate this diversity, bi-objective optimization has been proposed to simultaneously address the two conflicting objectives. This approach yields a Pareto front, which consists of a set of non-dominated solutions that reflect different investment strategies and provide investors with a comprehensive and diversified set of options. To guide this optimization process more effectively, we incorporate the trend ratio (TR) model [16], [17], which evaluates a portfolio's expected growth based on its deviation from a trend line. Portfolios exhibiting consistent alignment with the trend line are considered to carry lower risk, while greater deviations are interpreted as signals of instability. This makes the TR model particularly suitable for capturing the temporal dynamics of return and risk, providing a more nuanced and data-driven measure to steer bi-objective optimization in real-world markets.

To obtain a wide range of trade-off solutions, researchers have proposed various multi-objective optimization (MOO) methods, such as weighted sum approaches [18], ϵ-constraint methods [19], and a variety of evolutionary algorithms including NSGA-II [20]. However, as financial markets become increasingly dynamic and complex, more advanced algorithms are required to effectively capture asset interactions and generate high-quality, diverse solutions under multiple constraints. In this context, QA has demonstrated strong performance in single-objective optimization tasks, suggesting that it holds

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meaningful potential when extended to multi-objective portfolio optimization.

To our knowledge, this is among the first QA-based frameworks applied to bi-objective portfolio optimization, integrating the Trend Ratio (TR) model with D-Wave's constrained quadratic model (CQM) hybrid solver. The proposed method accounts for investor preference diversity, asset interactions, and cross-market dynamics by incorporating two distinct international stock indices, namely the Dow Jones Industrial Average (DJIA) from the United States and the Nikkei from Japan. By employing a scalarization-based approach with varying trade-off parameters, the framework generates a diverse set of non-dominated portfolios that reflect different investment priorities. The approach is evaluated using real-world financial data and is compared against classical MOO algorithms. In addition, the study investigates the impact of combining data from multiple markets to assess the effectiveness of cross-market diversification. The results confirm the feasibility and competitiveness of QA in bi-objective financial optimization and demonstrate its potential as a scalable solution for real-world investment decision-making.

II. Related Work

In this section, we review relevant literature from three distinct domains that help contextualize our work, including the application of QA to optimization problems, classical approaches for solving multi-objective optimization problems, and the use of the TR model in financial optimization.

QA [21] is a specialized quantum algorithm designed for solving combinatorial optimization problems. It has been widely applied across a range of industrial and commercial domains, including scheduling [22], [23], routing [24], [25], air traffic management [26], and machine learning [27]. To suit the architecture of quantum annealers, optimization problems are typically reformulated into quadratic unconstrained binary optimization (QUBO) problems or Ising models.

To formulate the bi-objective weighted portfolio optimization problem as a corresponding quadratic model, the QUBO framework has been the most commonly used approach in prior studies [28], [29]. This approach integrates multiple objectives and constraints, such as return and risk, into a single unconstrained binary optimization problem by introducing penalty terms. However, when dealing with large-scale problems or highly complex constraints, the design and tuning of penalty weights can become cumbersome and often result in reduced solution feasibility or accuracy. In contrast, the CQM offers greater flexibility, as it allows constraints to be expressed directly using equalities and inequalities, without the need to excessively rely on penalty terms. CQM supports up to 5,000 variables and 100,000 constraints, making it particularly suitable for solving large-scale optimization problems with complex constraint structures.

Moreover, most current applications of QA still focus on single-objective optimization, with relatively limited efforts directed toward addressing scenarios involving the simultaneous optimization of multiple potentially conflicting objectives. This emphasis on single-objective formulations largely stems from the inherent nature of QA, which operates by searching for the ground state of a Hamiltonian, an approach that naturally corresponds to minimizing a single objective function. As a result, when dealing with MOO problems, techniques such as scalarization are typically employed to combine multiple objective functions into a single solvable formulation or to convert secondary objectives into constraints. Currently, research on applying QA to MOO remains relatively limited [30], [31], indicating that this area still holds considerable potential for further exploration.

In contrast to the single-objective focus of QA, classical MOO techniques offer a well-established framework for handling problems with multiple conflicting goals. These methods aim to construct a Pareto front, providing a solution set in which no solution is superior to another in all objectives, thus allowing selection based on individual preferences. Classical MOO techniques include the weighted sum methods [18], ϵ-constraint methods [19], and multi-objective evolutionary algorithms (MOEAs) such as SMS-EMOA [32], MOEA/D [33], and NSGA-II [20]. SMS-EMOA adopts a hypervolume selection mechanism that promotes convergence and diversity by preserving individuals contributing most to the dominated hypervolume, making it particularly effective for problems with two or three objectives. MOEA/D adopts a decomposition strategy that transforms a multi-objective problem into multiple scalar subproblems, which are optimized simultaneously using an evolutionary algorithm. In contrast, NSGA-II combines fast non-dominated sorting with crowding distance sorting and an elitist strategy to balance convergence and diversity and is also effective in handling constrained multi-objective problems.

Despite the effectiveness of classical MOO techniques, applying them to real-world portfolio optimization remains challenging, especially in high-dimensional cases or when dealing with complex investment constraints. In this study, we consider a bi-objective portfolio optimization problem that involves financial markets from both the US and Japan. Incorporating these two distinct markets significantly enlarges the search space and increases the complexity of the problem. This added complexity arises from the need to capture the individual characteristics of each market along with their interactions, both of which affect the overall risk-return dynamics. These challenges illustrate the limitations of existing approaches and emphasize the need for new computational frameworks. Moreover, the application of quantum computing, particularly quantum annealing, to multi-objective financial decision-making is still in its early stages, yet holds substantial promise for future advancements in this field.

Building on this motivation, we incorporate the TR, a recently proposed performance metric that quantitatively evaluates a portfolio by integrating its expected daily return and daily risk into a single, unified measure. This formulation enables a more accurate, nuanced, and comprehensive assessment of investment performance over time, as it effectively balances the trade-off between return and risk. Prior studies

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[14] have shown that QA is effective in solving portfolio optimization problems formulated using the TR model, as it can efficiently navigate complex solution landscapes and identify portfolios that achieve stable growth while controlling risk. Based on these advancements, the present study extends the application of the TR model by incorporating it into a bi-objective portfolio optimization framework. In summary, while classical MOO methods have advanced significantly, their limitations in handling high-dimensional, constrained portfolio problems remain. By leveraging the unique capabilities of QA and integrating the TR model, this study presents a novel framework for bi-objective weighted portfolio optimization across a cross-market.

III. Proposed Method

A. Bi-objective Portfolio Optimization

Bi-objective optimization refers to optimization problems where two objectives must be optimized simultaneously, often with conflicting goals. In the context of portfolio optimization, these objectives are typically maximizing return and minimizing risk, which are inherently conflicting, as higher returns often come with higher risks. The following are some fundamental concepts related to bi-objective portfolio optimization, which are utilized in our proposed method.

1) Bi-objective Model: In a bi-objective optimization model, the two objectives are typically represented as a vector of functions, where each function corresponds to one objective, as shown in Eq. (1).

$$\text{Optimize } F(x) = [f_1(x), f_2(x)] \tag{1}$$

Here, $F(x)$ is the vector of objectives, and $f_1(x)$ and $f_2(x)$ represent the individual objective functions we aim to optimize. The goal of this problem is to find the values of x that optimize both $f_1(x)$ and $f_2(x)$, simultaneously considering the trade-offs between the two objectives.

2) Dominance: To evaluate the quality of solutions, a concept known as dominance is commonly used. In Eq. (2), with risk ($f_1(x)$) and return ($f_2(x)$) as objectives, we say that solution a dominates another solution b ($a \succ b$) if a has lower or equal risk ($f_1(a) \leq f_1(b)$) and higher or equal return ($f_2(a) \geq f_2(b)$) compared to b, and a has strictly lower risk ($f_1(a) < f_1(b)$) or strictly higher return ($f_2(a) > f_2(b)$). Conversely, b is said to be dominated by a because it is either riskier with the same or lower return, or less profitable with the same or higher risk.

$$\begin{aligned} a \succ b \iff & f_1(a) \leq f_1(b) \wedge f_2(a) \geq f_2(b) \\ & \wedge (f_1(a) < f_1(b) \vee f_2(a) > f_2(b)) \end{aligned} \tag{2}$$

Solutions that are not dominated by any other solutions form the non-dominated set (NDS), which represents the best trade-offs between risk and return. The NDS is identified as the Pareto front when all the solutions on it dominate all other solutions in both objectives.

3) Scalarization: Scalarization is the process of converting a bi-objective problem into a single-objective optimization problem, which is done by combining the objectives into a single function that reflects their relative importance.

A common scalarization technique is the weighted sum method, where the objectives are combined into a single objective function by assigning a weight to each objective, as shown in Eq. (3).

$$E(x) = \omega_1 \times f_1(x) + \omega_2 \times f_2(x) \tag{3}$$

In this equation, $E(x)$ represents the composite objective function that combines the two conflicting objectives $f_1(x)$ and $f_2(x)$. The values ω_1 and ω_2 are the weights assigned to each objective. These weights are non-negative and must satisfy the condition that $\omega_1 + \omega_2 = 1$, ensuring that the two objectives contribute proportionally to the composite objective. By varying the values of ω_1 and ω_2, the relative importance of each objective is determined, enabling the exploration of a range of solutions that balance the conflicting goals.

B. Weighted Portfolio Strategy

This section presents the stock allocation method employed in the proposed approach. Conventional fund allocation methods, particularly those constrained by QUBO, typically use binary variables (0 or 1) to determine whether each stock is selected, followed by an equal distribution of funds among the selected stocks. However, with the use of CQM, integer variables are allowed, enabling our method to explore more fund allocation combinations. In our method, each stock is assigned an integer value ranging from 0 to 100, representing the percentage of the initial funds allocated to each stock, weighted as x_i, the decision variable, as shown in Eq. (4), where N denotes the total number of stocks.

$$x_i \in \{0, 1, 2, \ldots, 100\}, \quad i \in \{1, 2, \ldots, N\} \tag{4}$$

In assigning x_i values to each stock, it is essential to ensure that the total investment does not surpass the initial capital, as specified in Eq. (5).

$$\sum_{i=1}^{N} x_i \leq 100 \tag{5}$$

The number of shares that can be purchased for each stock is then calculated, as shown in Eq. (6). ω_i is a function of x_i, representing how many shares can be purchased with the allocated funds for each stock, and is an essential part of the trend ratio calculation.

$$\omega_i = x_i \times \left\lfloor \frac{1\% \times funds_{ini}}{price_i(1)} \right\rfloor \tag{6}$$

Originally, ω_i was calculated by dividing the allocated funds by the stock price on the first day ($price_i(1)$), which gave the number of shares that could be purchased. However, due to rounding down the number of shares, we cannot directly

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divide the allocated fund, x_i, because of the calculation restriction imposed by the decision variables. Therefore, instead of directly dividing by x_i, we first calculate how many shares can be bought with 1% of the total funds ($funds_{ini}$) across all stocks, and then multiply the 1% stock share by the percentage allocated to each stock, thus achieving the same goal of the stock allocation while adhering to the restriction of CQM.

C. Trend Ratio Formulation for CQM Framework

In our proposed method, the TR model is used to evaluate the quality of each solution, presenting the increment ratio of each portfolio. The TR model focuses on two objectives: daily expected return (DER) and daily risk (DR), as shown in Eq. (7). However, due to the calculation restrictions of QA, the TR model must be reformulated into a form that can operate within the CQM framework. Eq. (8) refers to the TR under the CQM framework by treating DR as a positive term and DER as a negative term. The CQM solver minimizes this objective function, aiming to find the best portfolios with ω values that simultaneously consider minimizing DR and maximizing DER, thereby seeking solutions with the highest TR.

$$TR = \frac{DER}{DR} \tag{7}$$

$$E(\omega) = DR - DER \tag{8}$$

The DER and DR are primarily defined by the following equations, which have been modified to accommodate the computational limitations of the CQM framework.

$$FS(j) = funds_{ini} + \sum_{i=1}^{N} \omega_i \times (price_i(j) - price_i(1)) \tag{9}$$

$$DER = \frac{\sum_{j=1}^{D} (FS(j) - funds_{ini}) \times j}{\sum_{j=1}^{D} j^2} \tag{10}$$

$$TL(j) = DER \times j + funds_{ini},\ j \in [1, D] \tag{11}$$

$$DR = \frac{\sum_{j=1}^{D} (FS(j) - TL(j))^2}{D} \times \alpha \tag{12}$$

Eq. (9) defines the fund standardization $FS(j)$, which represents the portfolio value on day j. It is calculated using the initial fund value $funds_{ini}$, along with the portfolio weights ω_i and the changes in stock prices over time. Based on $FS(j)$, Eq. (10) calculates the daily expected return DER, which reflects the average rate of portfolio growth over a total of D days. Using DER, Eq. (11) generates a trend line $TL(j)$, which estimates the expected trajectory of the portfolio value over time. Finally, Eq. (12) measures the daily risk DR as the average squared deviation of the actual portfolio value from the expected trend, scaled by α. The value of α varies by experiment period, and further details are provided in [34].

D. Scalarization Weight Parameter λ

The previous section introduced a single-objective portfolio optimization method that maximizes the TR by treating DR and DER as equally important. In contrast, the bi-objective formulation considers that investors may have diverse preferences, with some emphasizing risk reduction and others pursuing higher returns. To address these variations, a set of non-dominated solutions is generated, each representing a different trade-off between risk and return.

To achieve this, we introduce a scalarization weight parameter λ to control a more precise trade-off between DER and DR, based on the weighted sum method. The key idea is to vary λ across multiple experiments, generating multiple solutions that reflect different preferences between the two objectives. The general equation is shown in Eq. (13).

$$E(\omega) = \lambda \times DR - (1 - \lambda) \times DER \tag{13}$$

The value of λ is within the interval $[0, 1]$, and in our approach, it increases from 0 to 1. When $\lambda = 0$, the contribution of DR is nullified, placing full emphasis on optimizing DER, resulting in solutions that focus on maximizing return, regardless of the minimization of risk. As λ increases, the solver will gradually focus more on DR, while the emphasis on DER will continuously decrease. When $\lambda = 0.5$, the solver gives equal weight to both objectives, leading to a balance between return and risk. Finally, when $\lambda = 1$, the solver focuses entirely on minimizing DR, regardless of DER maximization. By setting different λ, the model explores different maximum and minimum priorities between DER and DR, effectively generating diverse solutions.

To ensure a consistent variation of the λ value, we define a related parameter $\Delta\lambda$, which specifies the fixed increment of λ in each experiment, thus achieving equally weighted scalarization across all experiments. The relationship between λ and $\Delta\lambda$ is shown in Eq. (14).

$$\lambda_k = k \cdot \Delta\lambda \quad (k = 0, 1, 2, \ldots, \frac{1}{\Delta\lambda}) \tag{14}$$

In this formulation, the λ values are the multiples of $\Delta\lambda$, increasing until λ reaches 1. For example, if $\Delta\lambda = 0.2$, then λ will take values in the sequence {0, 0.2, 0.4, 0.6, 0.8, 1.0}, resulting in a total of six optimization runs, each generating solutions that focus on different risk and return priorities. Throughout these optimization processes, the encountered non-dominated solutions are retained. By aggregating them, a comprehensive non-dominated solution set is formed to represent the efficient frontier of both objective, thereby addressing the bi-objective portfolio optimization problem.

IV. Experimental Results

A. Environment Setting

This subsection outlines the experimental setup. The dataset covers three scenarios: the Dow Jones Industrial Average (DJIA) in the United States, which includes all 30 constituent stocks; the Nikkei 225 (NIKKEI) in Japan, from which the

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top 30 stocks by market capitalization are selected; and a cross-market scenario combining stocks from both indices, resulting in a total of 60 assets for evaluating international diversification. All stock data are obtained from Wharton Research Data Services (WRDS). The experiment period was set from 2022 to 2024, covering a total of three years. This period consists of 36 monthly intervals, providing a recent representative dataset for evaluation. The solver for QA is D-Wave's Leap hybrid CQM solver, which operates on the Advantage™ quantum system. Each QA run was assigned a time limit of 5 seconds, adhering to the solver's minimum required runtime to ensure adequate problem initialization and processing. The $\Delta\lambda$ value used in the first two experiments is 0.1. For algorithm comparison, the cross-market scenario is selected. SMS-EMOA and NSGA-II are both configured with a population size of 100, 250 generations, a crossover probability of 0.9, and a mutation probability of $\frac{1}{60}$, corresponding to the reciprocal of the number of stocks. MOEA/D adopts the same settings, with a neighborhood size T set to 10.

B. Performance Metrics

This study evaluates the quality and diversity of the obtained NDS using performance metrics, including Inverted Generational Distance (IGD), Hypervolume (HV), and the HV rate. However, IGD calculation requires the true Pareto front, but it is difficult to obtain due to the high complexity of the weighted portfolio optimization space. To address this, we construct a reference set called the combined NDS, which simulates the true Pareto front with its ideal solution quality and enables consistent evaluation of different solution sets. It is formed by performing non-dominated sorting over all available solutions from different algorithms and configurations, ensuring that only globally non-dominated solutions are included. The combined NDS is illustrated by the yellow curves in Fig. 1, and is defined as follows:

$$P(Y) = \{y \in Y \mid \nexists y' \in Y,\ f(y') \succ f(y)\} \tag{15}$$

where Y is the union of solution sets from all tested methods and configurations. A solution y is included in the combined NDS if and only if there exists no other solution $y' \in Y$ that dominates y. This combined set serves as a unified reference front, enabling objective and consistent evaluation of all solution sets using IGD and HV metrics.

1) Inverted Generational Distance (IGD): The IGD, as defined in Eq. (16), measures how well the obtained solutions approximate the true Pareto front, which is approximated by the combined NDS in this study.

$$\mathrm{IGD}(P, Q) = \frac{1}{|P|} \sum_{v \in P} d(v, Q) \tag{16}$$

A lower IGD indicates closer proximity to the ideal NDS, demonstrating better convergence and distribution of solutions. In this study, return and risk values were normalized to [0,1] prior to IGD calculation for consistency across experiments.

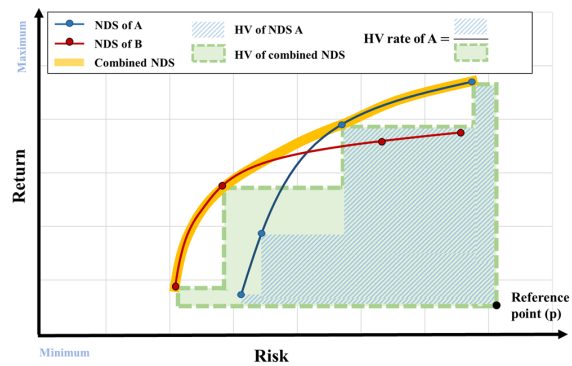

Fig. 1: The concept of the combined NDS is based on connecting the non-dominated solutions among the NDS of A with those from the NDS of B.

2) Hypervolume (HV): The HV metric measures the volume dominated by the NDS within the objective space, bounded by a reference point, and is defined as Eq. (17).

$$\mathrm{HV}(S) = \bigcup_{x \in S} \prod_{i=1}^{n} |f_i(x) - p_i| \tag{17}$$

A higher HV indicates better coverage and trade-off quality. In this study, HV is normalized, with the reference point set as the highest profit multiplied by 1.01 and the lowest risk multiplied by 0.99, excluding the point $(0, 0)$ to encourage competition. To ensure consistency, the HV rate is introduced, defined as the HV of the obtained NDS divided by the HV of the combined NDS. Fig. 1 illustrates this as the ratio between the blue (NDS A) and green (combined NDS) areas, reflecting coverage of the optimal trade-off space.

C. Evaluating the Performance of the Proposed Method in Cross-market

In the first experiment, we apply our model to the cross-market, aiming to test the model's performance as the solution space increases, while also observing the potential interactions between stock markets of different countries. Attempting to combine stock markets is a promising approach, as it can reveal more portfolio options than analyzing each market individually, due to the expansion of stock choices and the non-conflicting market environments. This supports our goal of meeting preference-based investment needs, thereby expanding the applicability of the proposed method.

Table I shows the average performance metrics over the three years of data, with the cross-market achieving the best results among all markets. This indicates that it can offer better portfolio choices in both quality and diversity than individual markets, offering a promising environment for investment decisions and highlighting the value of applying the proposed method to portfolio optimization in cross-market.

Fig. 2(a) represents the plot of three different markets in October 2022. During this month, the DJIA consistently outperforms the NIKKEI portfolios when the risk level is below approximately 150,000. However, beyond this value,

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TABLE I: Overall performance comparison across individual and cross-market portfolios, with the cross-market achieving the best results.

Market	HV	HV Rate	IGD
DJIA	0.570	76.40%	9.73E-02
NIKKEI	0.568	76.98%	9.88E-02
Cross-market	**0.725**	**98.34%**	**5.63E-03**

the DJIA portfolio gradually begins to be dominated. This presents an interesting phenomenon in which, if investors are limited to choosing only one market, they might prefer the DJIA if their risk tolerance is below a certain threshold, or the NIKKEI if they prioritize return and can tolerate higher risk. In contrast, by investing in the cross-market, both types of investors can freely select their preferred portfolios, as it not only incorporates these benefits but also offers a wider range of portfolios with better risk-return trade-offs, making the overall solution more optimal.

Similar behavior is observed in Fig. 2(b), where the cross-market is able to leverage the NIKKEI, creating a superior set even though the NIKKEI appears to be dominated in all solutions. This suggests that, regardless of how individual markets behave, the cross-market holds the advantages of both and can offer better portfolio options, further validating the potential of utilizing multiple markets together.

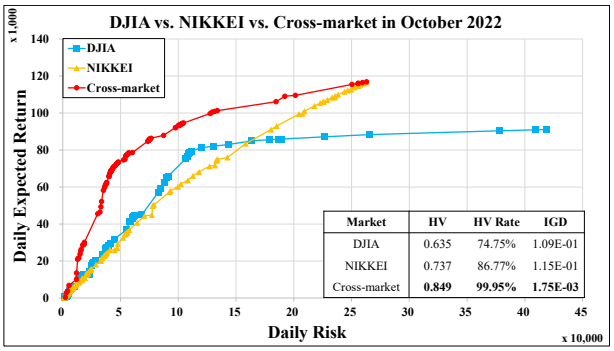
D. Comparison with MOEAs in Cross-market

In this section, we compare the solutions generated by the proposed method with those obtained from MOEAs, including SMS-EMOA, NSGA-II, and MOEA/D. The objective is to evaluate the solution quality and performance of each algorithm in the cross-market, thereby reinforcing the validity of our approach and demonstrating its competitiveness compared to conventional MOO algorithms in solving real-world cross-market portfolio optimization problems.

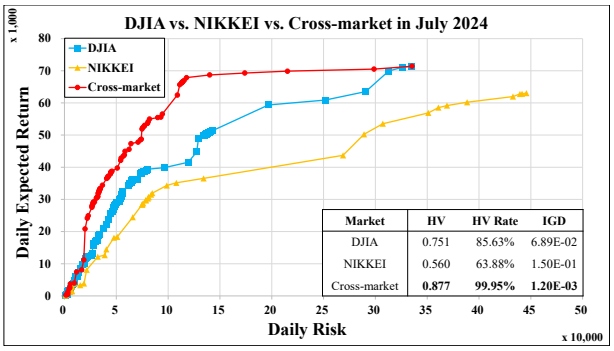
Table II presents the average performance metrics of each algorithm during the experimental period in the cross-market. The proposed method achieves the best overall performance, with the highest HV and HV Rate, the lowest IGD, and the greatest number of non-dominated solutions, offering a wide range of valuable investment options.

Fig. 3 provides further insight into the performance of each algorithm. As illustrated in Fig. 3(a), the non-dominated solutions produced by the conventional algorithms are highly competitive, particularly under the challenging 60-stock cross-market scenario. Each algorithm demonstrates strengths in different risk regions. Some favor low-risk portfolios while others perform better in high-return areas, resulting in a braided and intertwined Pareto front that is characteristic of complex multi-objective optimization problems. Despite this, our proposed method consistently delivers a diverse set of high-quality solutions, underscoring its robustness and adaptability across varying market conditions.

In bi-objective portfolio optimization, finding solutions in the low-risk domain is challenging due to the increased complexity, requiring finer adjustments to achieve optimal



(a) Comparison of different stock market in October 2022



(b) Comparison of different stock market in July 2024

Fig. 2: The visualization of the solution quality for DJIA, NIKKEI, and the cross-market shows that regardless of how individual markets behave, the cross-market yields more suitable portfolio options.

TABLE II: The comparison between our method and MOEAs shows that our method achieves higher quality and diversity of solutions without incurring excessively long runtime.

Algorithm	HV	HV rate	IGD	NDS#	Time(s)
SMS-EMOA	0.658	92.30%	5.55E-02	54	1.362
NSGA-II	0.660	92.55%	5.21E-02	50	0.305
MOEA/D	0.659	92.32%	4.93E-02	49	7.433
Our Method	**0.697**	**98.02%**	**4.76E-03**	**92**	0.337

trade-offs between risk and return. Fig. 3(b) shows that our method can find numerous low-risk solutions, distinguishing them from those generated by conventional algorithms, and provides a wide range of low-risk preference solutions, adding diversity to the overall investment choices.

In conclusion, our approach delivers high-quality and diverse solutions, providing several optimal portfolio options and addressing the gap where solutions in the low-risk domain are difficult to find, making it a suitable algorithm for solving bi-objective portfolio optimization in the cross-market.

E. Self-Analysis of the parameter $\Delta\lambda$

In the proposed method, λ is a scalarization weight that plays a crucial role in the optimization process, and it is primarily influenced by the value of $\Delta\lambda$. It not only af-

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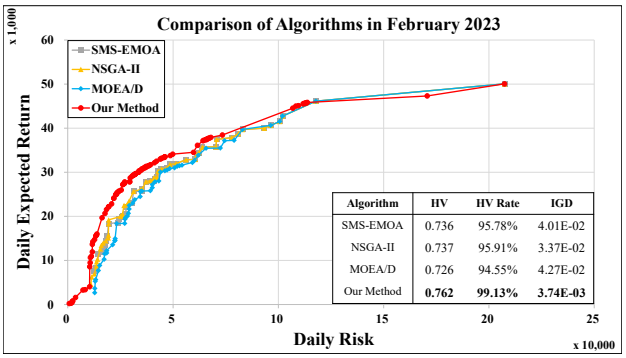
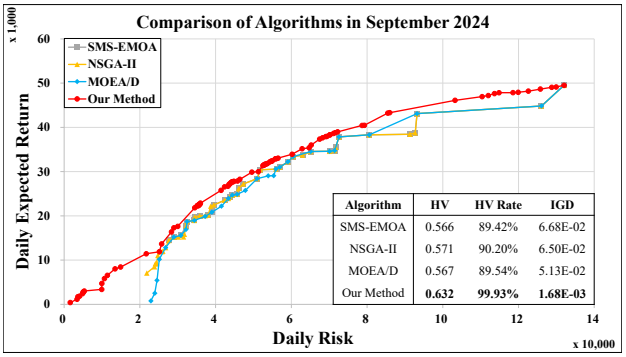


(a) Comparison of different algorithms in February 2023



(b) Comparison of different algorithms in September 2024

Fig. 3: The plot of different solution sets among the algorithms shows the capability of the proposed method to generate competitive sets compared to conventional evolutionary algorithms in the cross-market.

fects the number of solutions generated and the precision in capturing the trade-off between the two objectives, but also influences the number of experiments, thereby impacting the overall runtime. Therefore, the value of $\Delta\lambda$ must be scaled appropriately to maintain runtime efficiency. In this section, we present several experiment results for different $\Delta\lambda$ values and discuss which value of $\Delta\lambda$ is most appropriate for solving multi-objective portfolio optimization.

As $\Delta\lambda$ gets smaller, it should produce higher quality solutions due to the more precise risk-return preference, and more solutions due to the increased number of experiments. Table III presents the average performance metrics over a three-year period across all three markets. As shown, smaller $\Delta\lambda$ values consistently result in higher HV, higher HV rate, and lower IGD, along with a greater number of solutions compared to larger $\Delta\lambda$ values. This indicates that the quality of the solution set does improve with smaller $\Delta\lambda$ values. However, as $\Delta\lambda$ decreases, the runtime grows almost exponentially.

This leads to another discussion, where although a smaller $\Delta\lambda$ generates better NDS, it raises the issue of how small it should be. Because QA resources are limited due to hardware restrictions in current technology, the relationship between runtime and solution quality is not on the same scale. Instead,

TABLE III: The result shows that with a lower $\Delta\lambda$ value, the quality of solutions increases, but this comes at the cost of increased runtime.

$\Delta\lambda$	HV	HV rate	IGD	NDS#	Time(s)
0.2	0.620	90.13%	2.60E-02	55	**0.188**
0.1	0.651	95.06%	1.21E-02	92	0.344
0.05	0.671	98.07%	5.01E-03	149	0.661
0.025	**0.682**	**99.91%**	**2.84E-04**	**236**	1.289

TABLE IV: As the $\Delta\lambda$ value decreases, the HV improvement rate also decreases, while the runtime growth increases more, making the overall efficiency less ideal than desired.

$\Delta\lambda$	HV improvement	Runtime growth rate
0.2	-	-
0.1	**5.121%**	**82.543%**
0.05	3.000%	92.048%
0.025	1.705%	95.059%

the improvement rate, along with runtime, should be given more consideration. Table IV presents the HV improvement rate and the runtime growth rate, comparing each $\Delta\lambda$ value with its larger value. As you can see, the HV improvement rate continuously decreases, while the runtime growth rate increases. Fig. 4 further presents the total percentage of HV improvement rate and runtime growth rate, compared to the $\Delta\lambda$ value of 0.2. This shows that although the quality of the solution improves as $\Delta\lambda$ gets smaller, the improvement is not proportional to the increase in runtime, suggesting that smaller $\Delta\lambda$ values may not always be the optimal choice.

The results indicate that while reducing $\Delta\lambda$ leads to marginal gains in solution quality, it also causes a substantial increase in computational time. Among the tested settings, $\Delta\lambda = 0.1$ strikes a reasonable balance between solution quality and runtime, providing a good trade-off without incurring excessive computational costs. This makes it a practical and effective choice for the proposed method in bi-objective portfolio optimization.

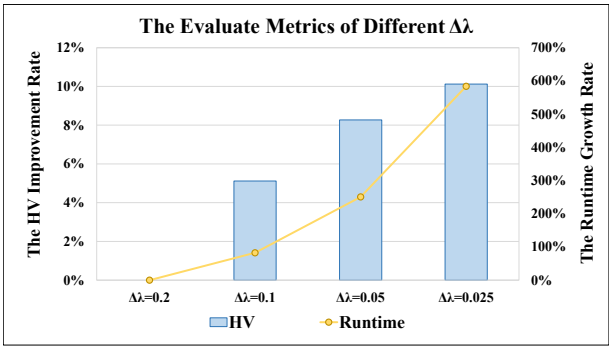


Fig. 4: The overall HV improvement rate and runtime growth rate under different $\Delta\lambda$ values show that while the runtime grows significantly, the improvement is only slight.

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V. Conclusion

This study presents the first application of QA to bi-objective weighted portfolio optimization by integrating the TR model with D-Wave's CQM hybrid solver across international stock markets. Leveraging scalarization, the proposed method generates diverse non-dominated portfolios that reflect varying investor preferences between risk and return. Experiments based on real-world financial data from the US and Japan explore the impact of cross-market integration on optimization outcomes. The results indicate that the method delivers high-quality and well-distributed solutions with favorable runtime characteristics. The cross-market setting further illustrates meaningful trade-offs, supporting the applicability of the approach to international investment scenarios. Comparative evaluations with NSGA-II, SMS-EMOA, and MOEA/D demonstrate that the method consistently yields strong performance, particularly in identifying low-risk portfolios and maintaining balanced outcomes across evaluation metrics. Analysis of the scalarization parameter $\Delta\lambda$ further suggests that moderate values (e.g., 0.1) offer a practical balance between solution quality and computational cost. Overall, the findings highlight the promising potential and practical effectiveness of QA and CQM-based hybrid solvers in addressing complex bi-objective weighted portfolio optimization.

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